

Spectral-Spatial Hyperspectral Image Classification: A Comparative Evaluation of Support Vector Machine, 2D and 3D Convolutional Neural Networks

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ABSTRACT

Hyperspectral images contain detailed spectral information for every pixel in a scene, making them useful for distinguishing between materials that appear similar in conventional imagery. In this work, three classification approaches were evaluated on the Indian Pines dataset: PCA-SVM, CNN-2D, and CNN-3D. PCA was first used to reduce the spectral dimensionality from 200 bands to 30 principal components while preserving most of the useful information. The obtained results showed that CNN-2D performed better than the other evaluated approaches. During experimentation, the model achieved an accuracy of 96.63%, whereas CNN-3D and SVM achieved 91.90% and 90.24%, respectively.

Keywords: Hyperspectral image classification; Convolutional neural networks; Support vector machine; Principal component analysis; Remote sensing.

NOMENCLATURE

CNN	Convolutional Neural Network
CNN-2D	Two-dimensional CNN
CNN-3D	Three-dimensional CNN
HSI	Hyperspectral Image
OA	Overall Accuracy
PCA	Principal Component Analysis
RBF	Radial Basis Function
SVM	Support Vector Machine
AVIRIS	Airborne Visible/Infrared Imaging Spectrometer
F1	F1-Score

1. INTRODUCTION

Traditional RGB images capture information in only three colour channels. In contrast, hyperspectral images contain data from many spectral bands, allowing each pixel to carry much richer information about the materials present in the scene. Because of this capability, hyperspectral imaging has found applications in agriculture, environmental monitoring, mineral exploration, and defence surveillance.

In defence-related scenarios, hyperspectral sensors can assist in tasks such as terrain analysis, object identification, and detection of concealed targets. The additional spectral information often provides details that are difficult to obtain using conventional imaging systems.

The high dimensionality of HSI data introduces challenges collectively referred to as the curse of dimensionality. Classical approaches typically address this via linear dimensionality reduction — most

commonly PCA — followed by a discriminative classifier such as the SVM. Whilst this pipeline has demonstrated satisfactory performance in numerous benchmark studies, it relies on handcrafted feature representations and does not inherently exploit spatial relationships between neighbouring pixels.

Deep convolutional neural networks have substantially altered this landscape by learning hierarchical feature representations directly from labelled training data. Two-dimensional convolution applied to spatial patches enables contextual feature extraction, whilst three-dimensional convolution further captures inter-band spectral correlations within a unified volumetric framework. The objective of this work was to compare three commonly used hyperspectral classification approaches under the same experimental setup. By evaluating PCA-SVM, CNN-2D, and CNN-3D on the Indian Pines dataset, the study aims to understand how the inclusion of spatial information influences classification performance.

The work extends a prior comparative study conducted in earlier work by Salam and Saxena¹, wherein qualitative results were reported without precise numerical accuracy values and did not include a CNN-3D evaluation. The present investigation addresses these gaps through rigorous quantitative benchmarking, per-class analysis, and reproducible experimental protocols.

Related Work

Earlier studies have shown that SVM can be an effective classifier for hyperspectral imagery, especially when the available training data is limited. Researchers later introduced PCA-based preprocessing to reduce spectral redundancy and improve computational efficiency. With the advancement of deep learning, CNN-based approaches became popular because they can automatically learn useful spectral and spatial features directly from data.

Dataset

The Indian Pines dataset was selected because it is widely used in hyperspectral image classification research, making it easier to compare the obtained results with previous studies. The dataset was collected by the AVIRIS sensor over agricultural fields in Indiana, USA. After removing noisy and water-absorption bands, 200 useful spectral bands remain for

analysis. Ground-truth labels are available for 16 land-cover categories.

The dataset presents several well-documented classification challenges: severe class imbalance (sample counts per class range from 4 to 2,455 in the complete ground-truth map); spectral confusion between agriculturally similar classes such as Corn-notill, Corn-mintill, and Corn; and the presence of mixed pixels at class boundaries. These characteristics make it a stringent and representative benchmark for evaluating classifier performance. Table 1 summarises the dataset properties alongside two additional validation datasets considered for future work.

Table 1. Hyperspectral benchmark datasets employed in this study

Dataset	Sensor	Spatial Size	Bands	Classes
Indian Pines	AVIRIS	145×145	200	16
Pavia University	ROSIS	610×340	103	9
Salinas Valley	AVIRIS	512×217	204	16

Methodology

Dimensionality Reduction via PCA

The 200-band hyperspectral image cube is reshaped into a two-dimensional matrix of dimensionality $N \times 200$, where N denotes the total pixel count. PCA was applied to reduce the number of spectral features while retaining most of the useful information present in the original data. After experimentation, 30 principal components were selected because they preserved more than 98% of the total variance.

Local Spatial Patch Extraction

A contextual patch of size $5 \times 5 \times 30$ is extracted around each labelled pixel, providing the classifier with local spectral-spatial neighbourhood information. Reflect-mode padding is applied to the spatial borders of the reduced image cube to ensure that boundary pixels yield complete patches. The total labelled sample pool is partitioned into training and test subsets at an 80:20 ratio using stratified random sampling to preserve class-proportional representation. The resulting test partition comprises 2,050 samples distributed across all 16 classes.

SVM Classifier

Each $5 \times 5 \times 30$ patch is flattened to a one-dimensional feature vector of length 750 prior to SVM input. A Radial Basis Function kernel is employed, with the regularisation parameter C and the kernel bandwidth γ optimised via five-fold cross-validated grid search on the training partition. Optimal hyperparameters were identified as $C = 10$ and $\gamma = \text{'scale'}$, the latter denoting normalisation by the product of feature dimensionality and training-set variance.

CNN-2D Architecture

The CNN-2D model processes each patch as a 5×5 spatial input with 30 spectral channels, analogous to a three-channel RGB image in conventional computer vision. The CNN-2D model receives a 5×5 spatial patch containing 30 spectral channels as input. Two convolutional layers are used to learn local spatial patterns from these patches. The extracted features are then passed through a fully connected layer before classification. Dropout was included during training to reduce overfitting and improve generalisation.

Table 2. CNN-2D layer configuration

Layer	Configuration	Output Shape
Input	$5 \times 5 \times 30$ patch	(5, 5, 30)
Conv2D	32 filters, 3×3 , ReLU	(3, 3, 32)
Conv2D	64 filters, 3×3 , ReLU	(1, 1, 64)
MaxPool2D	2×2	(1, 1, 64)
Flatten	—	(64,)
Dense	128 units, ReLU	(128,)
Dropout	rate = 0.40	(128,)
Dense (Out)	16 units, Softmax	(16,)

Training was conducted using the Adam optimiser with an initial learning rate of 1×10^{-3} , a mini-batch size of 64, over 10 training epochs. Data augmentation comprising horizontal and vertical reflections was applied to the training patches to reduce overfitting.

CNN-3D Architecture

For the CNN-3D model, each patch is reshaped to $5 \times 5 \times 30 \times 1$, introducing a unit channel dimension to accommodate three-dimensional convolutional kernels. Unlike CNN-2D, the CNN-3D model analyses spatial and spectral information together. The input patch is treated as a small volume, allowing three-dimensional convolutional kernels to capture relationships between neighbouring pixels as well as neighbouring spectral bands. This enables the model to

learn combined spectral-spatial features directly from the data.

Table 3. CNN-3D layer configuration

Layer	Configuration	Purpose
Input	(5, 5, 30, 1)	Spectral-spatial volume
Conv3D	32 filters, (3,3,7), ReLU	Volumetric feature extraction
Conv3D	64 filters, (3,3,5), ReLU	Deeper spectral-spatial features
MaxPool3D	(1, 1, 2)	Spectral-axis downsampling
Flatten+Dense	128 units, ReLU	Feature aggregation
Dropout	rate = 0.40	Regularisation
Dense (Out)	16 units, Softmax	Class probability output

Experimental Setup

All experiments were implemented in Python 3.10 using TensorFlow 2.x / Keras for the neural network models and scikit-learn for the SVM pipeline. All experiments were performed by the author during the internship using Python-based machine learning frameworks. Multiple trial runs were carried out before selecting the final hyperparameter settings used in this study.

Results and Discussion

Overall Performance Comparison

Table 4 summarises the overall classification metrics for the three evaluated models on the Indian Pines test partition. Among the three approaches evaluated, CNN-2D produced the best results on the Indian Pines dataset. It achieved an overall accuracy of 96.63%, outperforming both CNN-3D and SVM. One possible reason is that the spatial features captured by the 2D convolutional layers were sufficient for distinguishing most land-cover classes present in the dataset. Although CNN-3D was designed to learn both spectral and spatial information simultaneously, its performance remained below that of CNN-2D in the current experimental setup. The relatively small patch size may have limited the amount of contextual information available to the network.

Table 4. Classification performance on Indian Pines test set (2,050 samples, 16 classes)

Model	OA (%)	Precision	Recall	F1
SVM (PCA+RBF)	90.24	—	—	—
CNN-2D	96.63	0.97	0.98	0.97
CNN-3D	91.90	0.94	0.91	0.92

CNN-2D Training Dynamics

Figures 1 and 2 illustrate the training and validation accuracy and loss trajectories for the CNN-2D model over ten epochs. During training, the model showed rapid improvement in the first few epochs. After approximately three epochs, both training and validation accuracy began to stabilise. The final validation accuracy reached 96.63%, indicating that the model was able to generalise well to unseen samples. The validation loss similarly reduces from 0.62 at epoch zero to approximately 0.10 at epoch nine, confirming stable model convergence without divergence.

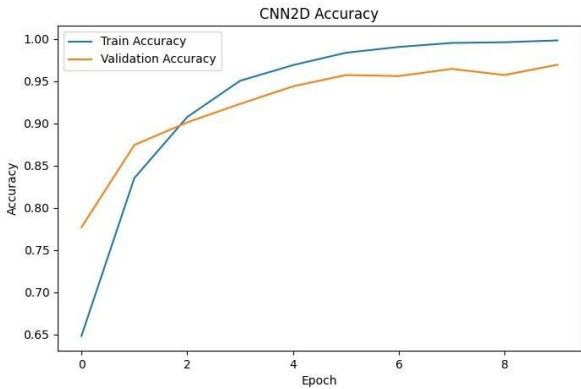


Fig. 1. CNN-2D training and validation accuracy over 10 epochs. Final validation accuracy: 96.63 per cent.

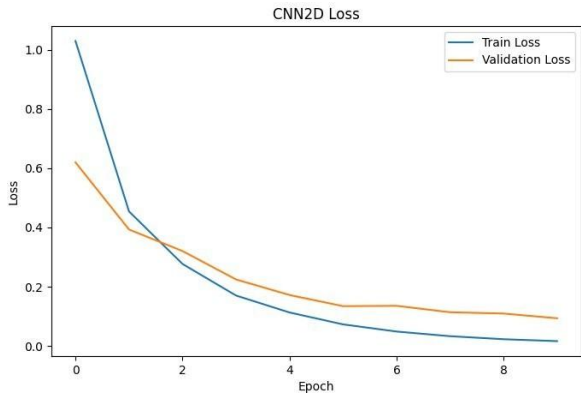


Fig. 2. CNN-2D training and validation loss over 10 epochs. Validation loss converges to approximately 0.10.

CNN-2D Confusion Matrix and Per-class Analysis

Figure 3 presents the confusion matrix for the CNN-2D classifier. Most classes were classified correctly,

which can be seen from the strong diagonal pattern in the confusion matrix. The majority of errors occurred between Corn-notill and Corn-mintill, and between Soybean-notill and Soybean-mintill. These classes have very similar spectral characteristics, making them difficult to separate.

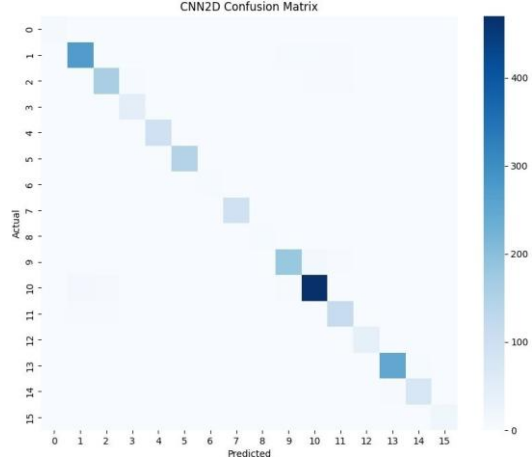


Fig. 3. CNN-2D confusion matrix on Indian Pines test partition (OA: 96.63 per cent). Class indices 0-15 correspond to the 16 land-cover categories.

CNN-3D Training Dynamics and Confusion Matrix

An interesting observation during CNN-3D training was that validation accuracy was initially higher than training accuracy. This behaviour is commonly observed when dropout regularisation is used. Although the model continued to improve throughout training, its final performance remained lower than CNN-2D. The validation accuracy converges to 91.90 per cent by epoch nine, whilst the validation loss stabilises at approximately 0.27, notably higher than the 0.10 recorded for CNN-2D. This elevated validation loss suggests that the volumetric kernels of CNN-3D encounter greater difficulty in discriminating between spectrally similar classes given the limited 5×5 spatial context available in the input patches.

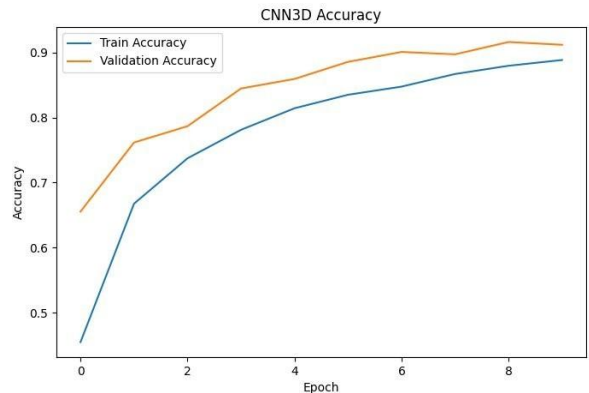


Fig. 4. CNN-3D training and validation accuracy over 10 epochs. Final validation accuracy: 91.90 per cent.

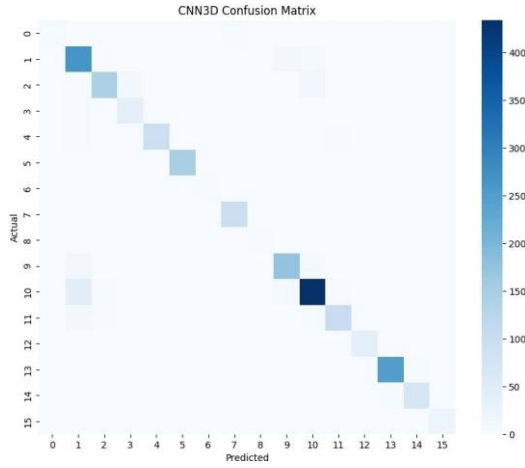


Fig. 5. CNN-3D confusion matrix on Indian Pines test partition (OA: 91.90 per cent). Class 0 (Alfalfa) exhibits reduced recall due to very limited test-set samples.

Analysis of the CNN-3D confusion matrix showed that the model struggled with a few minority classes. For example, Alfalfa contained very few test samples, which affected recall performance. Misclassifications were also observed among spectrally similar crop categories. These results indicate that CNN-3D was not able to utilise the limited 5×5 spatial context as effectively as CNN-2D.

Per-class F1-Score Comparison

Figure 6 provides a per-class F1-score comparison between CNN-2D and CNN-3D across all 16 Indian Pines classes. The per-class F1-score comparison further confirms the advantage of CNN-2D. Higher scores were obtained for most classes, particularly those involving spectrally similar crops. Only a few classes showed comparable performance between the two models. Overall, CNN-2D demonstrated better consistency across different land-cover categories.

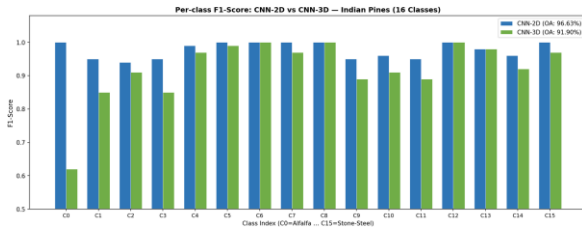


Fig. 6. Per-class F1-score comparison between CNN-2D and CNN-3D across 16 Indian Pines classes. CNN-2D (blue) consistently outperforms CNN-3D (green) on spectrally challenging classes.

Comparative Discussion

The SVM model achieves a competitive 90.24 per cent overall accuracy with substantially lower computational cost — training completes in seconds on CPU without GPU dependency, making it highly suitable for resource-constrained deployment

scenarios. However, the flattening of spatial patch structure prior to SVM input discards neighbourhood feature interactions that prove critical for discriminating spectrally similar classes.

Based on the obtained results, CNN-2D was the most effective approach for this dataset. The model benefited from spatial information present in neighbouring pixels, which helped distinguish classes that were spectrally similar. Although CNN-3D was expected to perform better because it uses both spectral and spatial information simultaneously, the limited patch size may have restricted its ability to fully utilise these advantages.

Compared with the earlier work by Salam and Saxena, the current implementation produced better classification performance. The use of an RBF kernel and systematic hyperparameter tuning improved SVM accuracy, while the inclusion of CNN-3D provided an additional benchmark for comparison. The present work also provides per-class precision, recall, and F1-score analysis, enabling a more detailed evaluation of classifier performance.

2. CONCLUSION

This work compared three different hyperspectral image classification approaches using the Indian Pines benchmark dataset. Experimental results demonstrated that CNN-based methods outperform the traditional PCA-SVM pipeline. Among all evaluated models, CNN-2D achieved the highest accuracy of 96.63%. The study also showed that the effectiveness of CNN-3D depends strongly on the available spatial context. Future work can explore larger patch sizes, additional benchmark datasets, and hybrid architectures that combine spectral and spatial learning more effectively.

The source code and trained models are publicly available at <https://github.com/KislayDubey29/DRDO-Hyperspectral-Classification>.

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CONTRIBUTORS

Kislay Dubey contributed to conceptualisation, software implementation, experimentation, data analysis, and manuscript preparation.

Raghvendra Sahai Saxena contributed to research conceptualisation, scientific supervision, and critical review of the manuscript.

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